Triple Balance - Background Information for Educators

1 The Environmental Dilemma

Saving the planet sounds great, right? But what if doing so drains the city's budget or makes its citizens so unhappy that they vote you out of office? Welcome to the environmental dilemma of Triple Balance! Smartphones may be small, but their environmental footprint is not. From mining precious raw materials to producing mountains of electronic waste, our love for smartphones comes at a cost. As mayor, your mission is to turn this throwaway culture into a circular economy where smartphones and their materials are used for as long as possible. But here is the dilemma: greener choices aren't always cheaper or more popular. Should you introduce stricter recycling rules, even if citizens complain? Invest in futuristic technologies, even if the city can barely afford them? Or encourage people to use their phones less, knowing they might not love the idea? Every decision comes with trade-offs, and there is no magic solution. The real challenge isn't just saving the environment, it's doing so without going broke or getting kicked out of office!

2 Overview and Learning Objectives/Outcomes

This activity involves a short digital game, "Triple Balance", which runs in any browser on computers, tablets, or even mobile phones. No installation needed.

https://imaginary.github.io/triple-balance-phone/

Through the lens of the mayor of a fictional city, the participants will make several decisions on the circular economy of smartphones. Examples include tax laws, recycling rules, a digital detox day, and the invention of new technologies (have you heard of 'smart bins' or a 'regenerative port matrix'?). There might also be sudden events the mayor needs to decide on, such as a fire at the recycling plant.

Even though some parts of the game seem purely fictional, there are some true parts about leading a city: The budget is limited, and if your decisions make the citizens really unhappy, you get thrown out of office. The goal of the game is to build a sustainable smartphone economy while balancing citizen happiness and the city budget (hence the name: Triple Balance).

By the end of the activity, participants will be able to:

- Identify and list at least three R-strategies (e.g., Reuse, Repair, Recycle).
- Predict which outcomes are more or less likely to occur when making decisions.
- Compare different decision paths and identify how they contribute to the variety of possible outcomes.
- Illustrate how a sequence of decisions can generate multiple possible outcomes.
- Explain the concept of the circular economy and describe key principles.
- Evaluate the effectiveness of different sustainability strategies.

The game is based on the multiplayer collaborative board game “Game of Phones”, which was also developed as part of this math4sustainability project:

https://math4sustainability.eu/r-strategies/

3 The Mathematics Behind It

The exhibit introduces participants to mathematical concepts related to probability, combinatorics, and decision-making.

Participants experience how decisions can lead to different outcomes, some of which are more likely to occur than others. Through gameplay, they develop an intuitive understanding of probability and uncertainty. On top of that, the game illustrates how a sequence of decisions can lead to a large number of possible combinations.

One playthrough is one journey from the start to an ending. Change a decision, get a different random event, or choose another response, and you follow a different path, even if you finish with the same score.

The set of all playthroughs can be described as a decision tree, where every branching point is a decision to be made. The total number of combinations gives us a sense of all possible outcomes of the game.

Let’s have a look at the choices in the game:

Policies

There are six rounds of policy decisions, each presenting a choice among three specific policies. So, the user will see a total of $3 \cdot 6 = 18$ different policies in the game, choosing up to 6 of them during a single playthrough. Note that a playthrough can end early if you run out of money or happiness.

Events

There are three stages during which events may occur. There are a total of 9 distinct events, and each can only happen in its predefined slot. In each of these event slots, the game picks one of three possible events, or no event (with evenly distributed probabilities). In case an event occurs, the player has to choose between two possible reactions. Thus, 7 possible outcomes for each event round. The player might be presented with up to three events in a game (3 slots).

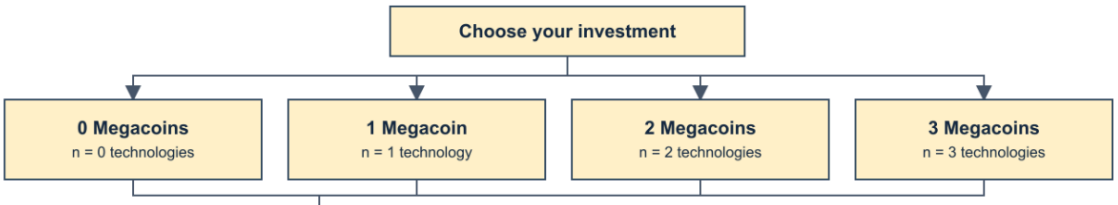
New Technologies

At the beginning of the game, you can choose to invest in research on new technologies, which give you up to 3 additional rounds of decisions. Each round presents the player with a choice among three specific technologies. Depending on the initial choice, the user will see 0, 3, 6, or 9 different new technologies in the game, choosing 0, 1, 2, or 3 of them during a single (successful) playthrough.

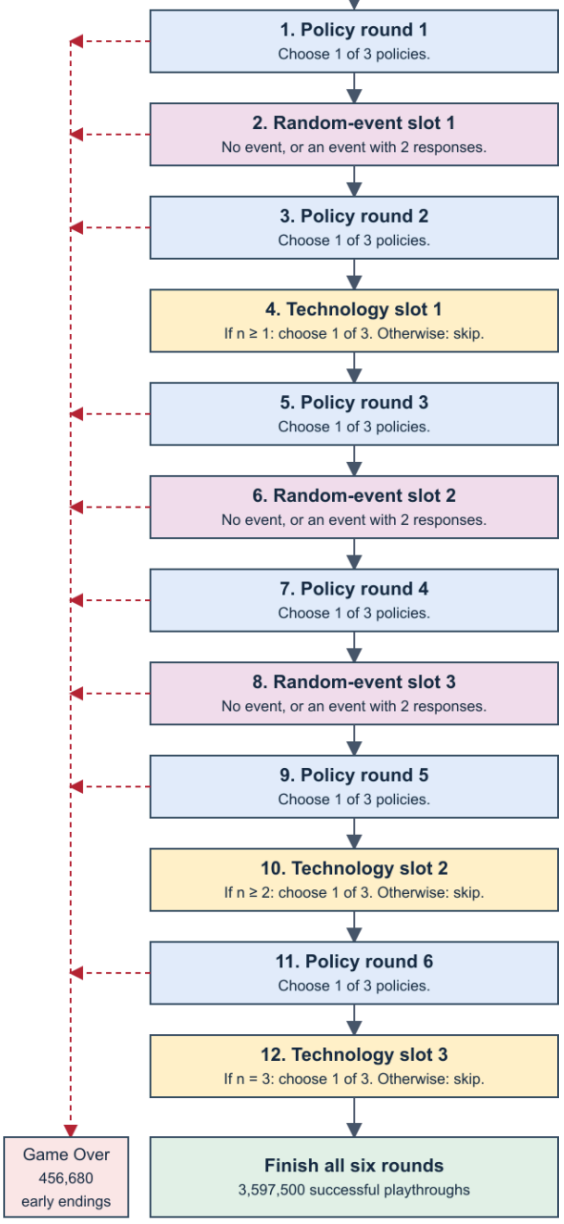
You can follow all the possibilities in this decision chart:

One game, millions of possible paths

1. Pay for research before the first policy round

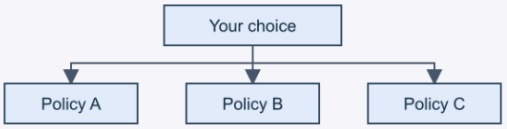


2. Follow the rounds



How the branches work

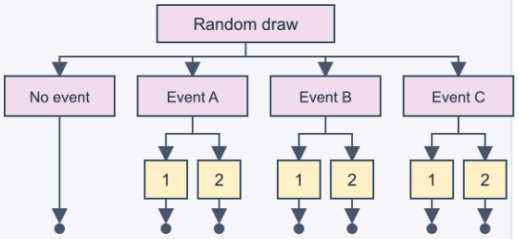
Each policy round: 3 branches



Each option continues with its own resources.

Each event slot: 1 + 3 × 2 = 7 branches

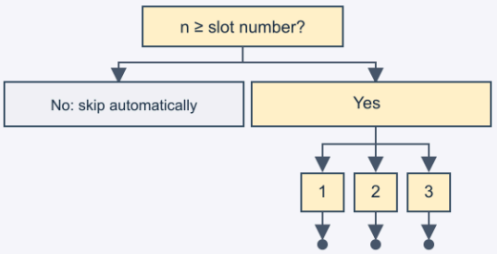
Each slot has its own events. You choose the response.



Response 1 or 2 for each event; no response for "no event".

These seven routes are not equally likely.

Technology slots depend on the initial payment



If n = 0, 1, 2, or 3, then thre are 1, 3, 9, or 27 branches, respectively.

Dashed red branches: stop at Game Over

After a policy or event response, stop if:

- happiness is zero or below, or
- the budget is zero or below.

Do not count any decisions after that point.

Playthroughs if we ignore Game Over: $3^6 \times 7^3 \times (1 + 3 + 9 + 27) = 10{,}001{,}880$

Actual playthroughs: 3,597,500 finishes + 456,680 early endings = 4,054,180

Count each visible path once. Exclude restarts and explanation/credits detours. Counts are not probabilities.

We can calculate all the possible path combinations disregarding whether the game ends early due to running out of money or happiness. That would be:

- 6 policy rounds with 3 outcomes on each round: 3^6 branches.
- 3 event rounds with 7 outcomes on each round: 7^3 branches.
- 1 branch with no technology, 3 branches with one technology, 9 branches with two technologies, and 27 branches with three technologies: 1+3+9+27 = 40 branches.

So the total number of combinations is

$$3^6 \cdot 7^3 \cdot (1 + 3 + 3^2 + 3^3) = 250047 \cdot 40 = 10\,001\,880.$$

That is more than ten million!

However, this is only an upper bound of the real playthroughs. The game stops when happiness or money falls to zero or below. Choices after that point do not count: many full plans share the same early ending. The real playthroughs can be split between those that lead to a sudden game over and those that successfully reach the end.

A computer program can follow every branch and stop counting that path at its ending. These are the results:

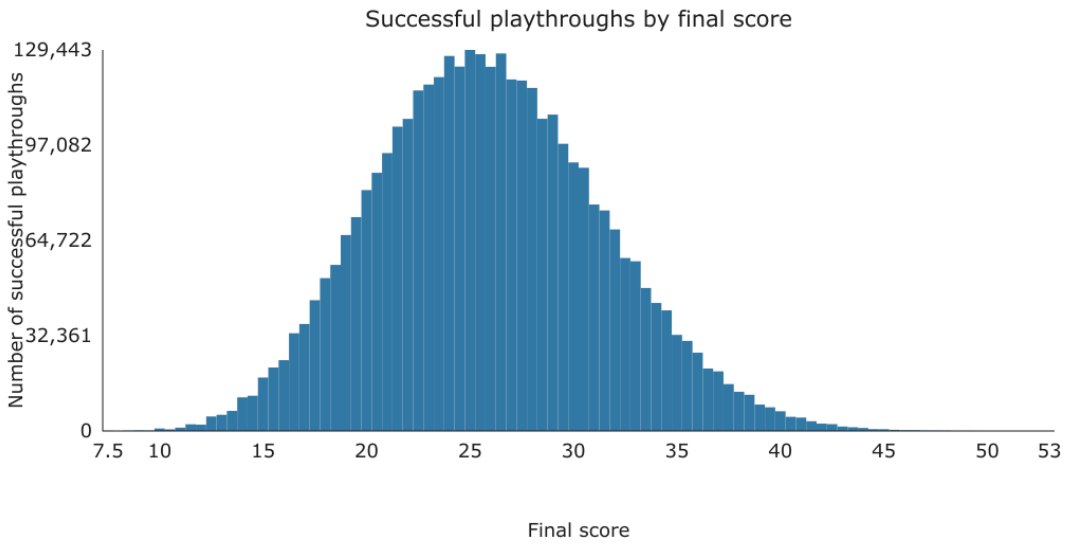
Finish all six rounds:	3 597 500
Game Over:	456 680
Total:	4 054 180

That is over four million different journeys! And yet, these are possible paths, not probabilities: the numbers do not tell us how often real players will win. We do not count restarts or visits to explanation and credits pages as additional branches.

If you succeed in finishing all six rounds, you are given a score. To calculate it, the values for circularity, happiness, and budget are taken into account. A weighted sum combines several criteria into a single score, giving some criteria more weight than others. For the game, circularity is the most important criterion. We also decided that happiness is more important than the leftover budget and came up with this formula for the score:

$$\text{Score} = 2.5 \cdot \text{Circularity} + 1.5 \cdot \text{Happiness} + 1 \cdot \text{Budget}$$

The program calculates the score distribution across all 3 597 500 successful playthroughs. That is, how many playthroughs end with 7.5 points, 8 points, 9 points, etc. (the minimum score for a successful playthrough is 7.5 and the maximum is 53). The program displays this distribution as a histogram.



This allows the program to give you a percentile score. If you get a percentile of X / 100, it means that X% of the possible playthroughs get a lower or equal score than you. For instance, a score of 37 out of 53 points would be at the 97th percentile, because 97% of possible playthroughs get less than or equal to 37 points. Note that 37/53 = 0.698 is not your percentile score. The median is the 50th percentile, that is a score such that 50% of games have a score lower than or equal to it. In this case, it is 25.5 (50.28% of the games score 25.5 or below). As we mentioned above, the count of playthroughs is not a probability. How likely a score is depends on your choices and your luck.

Conclusion

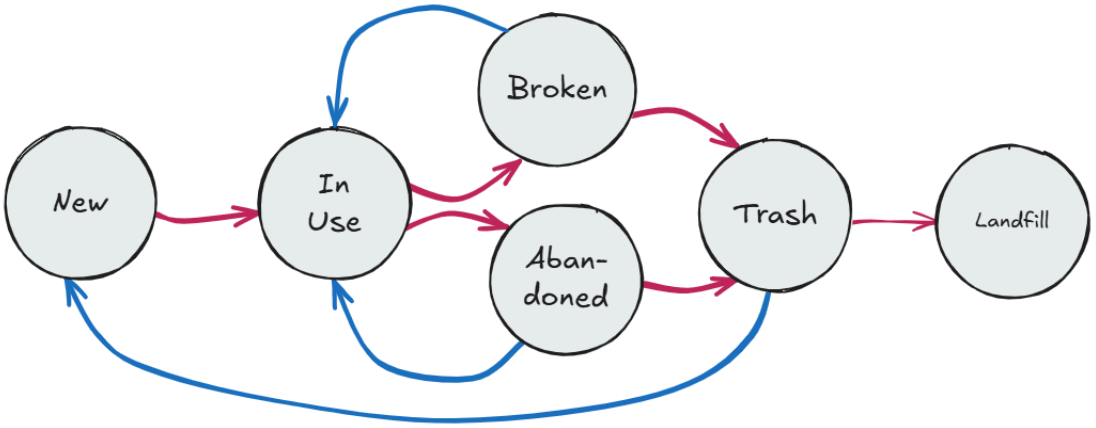
This math exercise is not an end in itself. By exploring probability, combinatorics, and decision-making, players discover how complex everyday decisions can be, especially when it comes to sustainability. With more than four million possible playthroughs, the game illustrates how quickly the number of possible outcomes grows as different choices and random events are combined. While mathematics can help us calculate, understand and compare these possibilities, the game also shows its limitations: random events and unforeseen consequences can disrupt even the best-laid plans, and every scoring system depends on which criteria we consider most important. When discussing environmental dilemmas, the challenge is therefore not simply to find the best outcome on paper, but to make informed decisions despite uncertainty and competing priorities.

4 Group Activities Step by Step

If you have played the boardgame *Game of Phones* https://math4sustainability.eu/r-strategies/, you can recall it in connection with this activity. If you want to play it, it would take about 1-2 hours, and we suggest playing the game on a different day than the rest of the activity.

Introduction (ca. 10 min)
Introduce the concept of circular economy using the example of smartphones.

- Brainstorm together on strategies for a circular economy, going through some of the R-strategies (e.g., Reduce, Reuse, Repair, Recycle).
- Compare it to a linear economy.
- Optional (additional 10 - 30 min, flexible):
  Jointly come up with a sketch of the linear economy of smartphones through brainstorming (only the stages and the red arrows in the sketch below).
  (For younger kids: prepare cards with the words beforehand and draw arrows on other cards for them to move around on a joint table or on the floor. You can also put them with magnets on a blackboard and draw the arrows with chalk.)
  Have participants come up with ideas about which backward-pointing arrows (the blue ones in the sketch) are possible and identify which R-strategy each represents.



The image depicts:

- Repair: Broken ⟶ In Use
- Reuse: Abandoned ⟶ In Use
- Recycle: Trash ⟶ New

Playtime (ca. 30 minutes, flexible)

Needed: computer, tablet, or phone
Participants use one device per two people (no more than three people, single-player is possible).

The game is self-explanatory and takes only a few minutes to complete a playthrough. Play several times. Encourage participants to achieve higher scores. You may collect high scores on the blackboard.

Discussion (optional)

The following questions are meant as examples to ignite discussion. The participants might have their own questions and points they want to discuss with the group. Always try to follow the lead of the participants.

- Did you have a strategy, or did you just make decisions as you went along?
- What did you find particularly hard?
- Was it possible to make everyone happy while also protecting the environment?
- Can you describe the dilemma you faced as the mayor of Phonopolis?
- Should environmental protection always come first, even if it costs a lot of money?
- Which situations in the game felt realistic? Which seemed completely fictional?
- Do you think real politicians face similar dilemmas?
- Would you rather be a popular mayor or a sustainable mayor?
- Would people support stricter recycling rules or a digital detox day in real life? Would you?
- Would you rather repair your phone or buy a new one? What would influence your decision?
- Would you buy a refurbished smartphone if it helped the environment?
- What matters more: developing new green technologies or consuming less in the first place?
- What's one thing you could change about your own smartphone habits to make them more sustainable?

<u>Math Challenge (optional)</u>

Participants should pair up in teams of 2 or 3.

Material:

- No device needed, just pen and paper, maybe a calculator.
- Print out the structure of the decision tree (version with no results, see link in the worksheet chapter 6.7) for each team. (see next page)

Go through the decision tree together, make sure everyone understands. Clarify questions as needed.

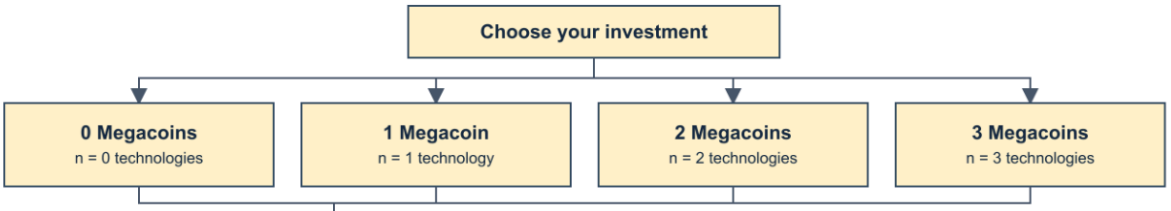
Task:
How many different combinations of decisions are theoretically possible?

Some specifications:

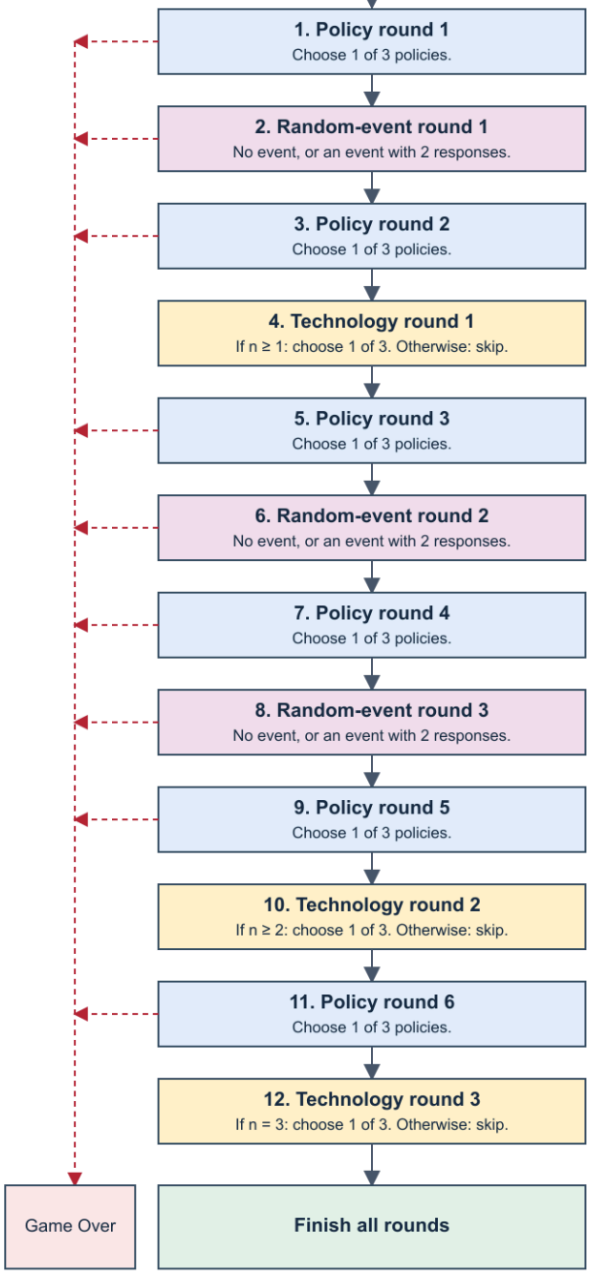
- Two combinations are different if the player makes at least one different choice.
- Ignore any early endings ("Game Over"). Assume that the game always continues through all 12 stages as shown in the diagram. Calculate the total number of theoretically possible combinations.

Depending on the participants' age group, you may ask only for the number of combinations, without considering any technologies.

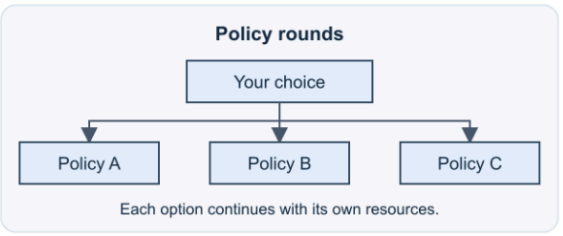
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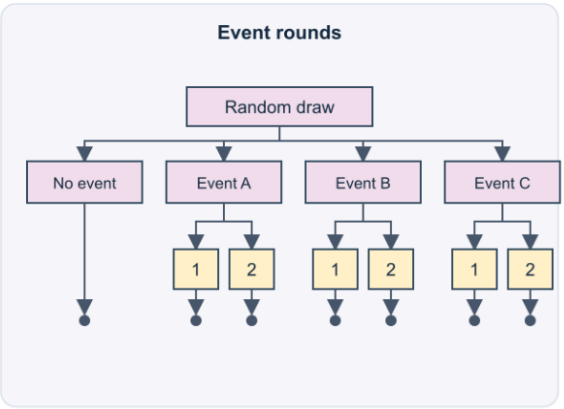


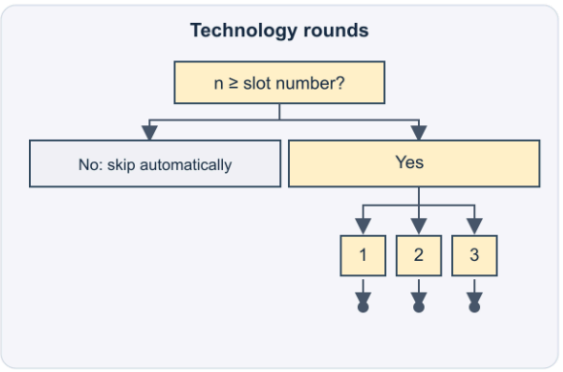
2. Follow the rounds



How the branches work







Dashed red branches: stop at Game Over

After a policy or event response, stop if:

- happiness is zero or below, or
- the budget is zero or below.

Do not count any decisions after that point.

Solution:

- So the total number of combinations is:
  $3^6 \cdot 7^3 \cdot (1 + 3 + 3^2 + 3^3) = 250047 \cdot 40 = 10\,001\,880$ (see the math chapter above)
- The number of combinations without any technologies is:
  $3^6 \cdot 7^3 \cdot 1 = 250\,047$

Just for fun (if time allows)

Material:

- Device to play the game
- Print-out of the decision overview (see link in the worksheet chapter 6.7), see also next page.

Participants can play the game again using their devices and a print-out of the decisions overview (Note: The overview is not a decision tree, it just lists all options). During each playthrough, they cross off the options they have chosen or encountered.

Goal: Try to cross off every option at least once across multiple playthroughs.

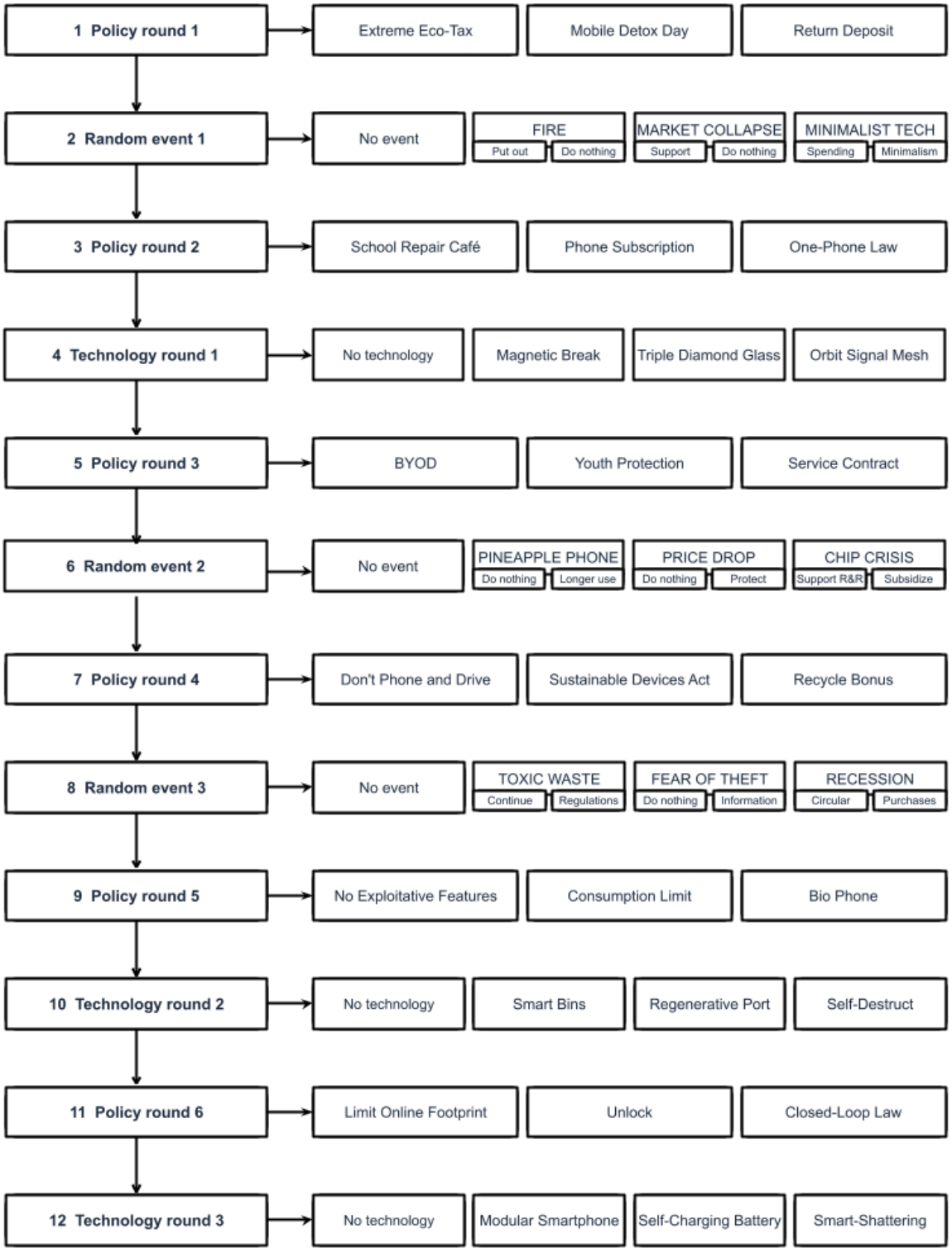
Question: **What is the minimum number of playthroughs needed to cover all options?**

Note: Events occur randomly, so encountering every possible event may take more time and patience than participants have. The activity can be stopped at any point.

- Policies: At least 3 playthroughs are needed to choose every policy at least once.
- Technologies: At least 4 playthroughs are needed to choose every option at least once, including ‘no technology’.
- Events: At least 7 playthroughs are needed to experience every event, both possible reactions to each event, and the ‘No event’ option. However, because events occur randomly, it may take considerably more playthroughs in practice.

These numbers represent the minimum number of playthroughs. They disregard early game terminations when the player runs out of money or happiness.

Key takeaway:
Just 7 playthroughs are theoretically enough to experience every option at least once. This is a striking contrast to the more than 10 million possible combinations!


1 Policy round 1
Extreme Eco-Tax
Mobile Detox Day
Return Deposit
2 Random event 1
No event
FIRE
Put out
Do nothing
MARKET COLLAPSE
Support
Do nothing
MINIMALIST TECH
Spending
Minimalism
3 Policy round 2
School Repair Café
Phone Subscription
One-Phone Law
4 Technology round 1
No technology
Magnetic Break
Triple Diamond Glass
Orbit Signal Mesh
5 Policy round 3
BYOD
Youth Protection
Service Contract
6 Random event 2
No event
PINEAPPLE PHONE
Do nothing
Longer use
PRICE DROP
Do nothing
Protect
CHIP CRISIS
Support R&R
Subsidize
7 Policy round 4
Don't Phone and Drive
Sustainable Devices Act
Recycle Bonus
8 Random event 3
No event
TOXIC WASTE
Continue
Regulations
FEAR OF THEFT
Do nothing
Information
RECESSION
Circular
Purchases
9 Policy round 5
No Exploitative Features
Consumption Limit
Bio Phone
10 Technology round 2
No technology
Smart Bins
Regenerative Port
Self-Destruct
11 Policy round 6
Limit Online Footprint
Unlock
Closed-Loop Law
12 Technology round 3
No technology
Modular Smartphone
Self-Charging Battery
Smart-Shattering

5 Curriculum Integration

Triple Balance has strong potential for integration into secondary education as an interactive learning activity that helps students explore the challenges of environmental sustainability and the transition to a circular economy. Using smartphones as a familiar example, the game introduces students to environmental problems associated with resource extraction, production, consumption, and electronic waste.

The game can be used at different stages of a learning unit. As an introduction, it can spark curiosity and raise awareness of the dilemmas associated with sustainable development. After students have learned about circular economy principles, it can provide an opportunity to apply their knowledge and evaluate different strategies.

The game also offers opportunities for interdisciplinary learning, particularly in mathematics. Using the game's decision tree, students can apply combinatorics.

France

Triple Balance can be integrated into lower-secondary education (collège), particularly in Geography, Moral and Civic Education (Enseignement moral et civique, EMC), Technology, and Mathematics. In Geography, the game connects to the sustainable management of natural resources, environmental challenges, and responsible consumption. In EMC, students explore political decision-making and the dilemmas that arise when environmental protection conflicts with economic interests or public acceptance. Technology lessons can use the game's smartphone theme to investigate product lifecycles, recycling, and sustainable technological solutions. In Mathematics, students can apply combinatorics and probability to calculate the number of possible playthroughs and examine the role of random events.

Germany

Triple Balance is particularly suitable for lower-secondary education (Sekundarstufe I) in Geography, Politics/Economics (Politik/Wirtschaft), Ethics, and Mathematics. The game supports the principles of Education for Sustainable Development (Bildung für nachhaltige Entwicklung, BNE), encouraging students to understand complex relationships, evaluate competing interests, and reflect on responsible decisions. Its focus on sustainability dilemmas allows students to explore conflicts between environmental protection, economic stability, and social acceptance. In mathematics, the game's decision tree provides a practical application of combinatorics and probability, allowing students to calculate theoretically possible playthroughs and investigate random events. Specific curriculum connections depend on the federal state (Bundesland).

The Netherlands

Triple Balance can be integrated into lower-secondary education (onderbouw voortgezet onderwijs), particularly in Geography (Aardrijkskunde), Citizenship Education (Burgerschap), Economics (Economie), and Mathematics (Wiskunde). The game supports learning about sustainable resource use, responsible consumption, democratic decision-making, and economic trade-offs. By acting as a mayor, students experience how environmental policies can conflict

with financial constraints and citizens' preferences. This makes the game especially suitable for developing critical thinking, argumentation, and an understanding of sustainability dilemmas. In Mathematics, students can explore combinatorics, probability, and decision trees by calculating possible playthroughs and examining the influence of random events.

Cyprus

Triple Balance is well suited for lower-secondary education (Gymnasium), particularly in Environmental Education and Education for Sustainable Development, Geography, Home Economics, and Mathematics. The game introduces students to the environmental impacts of smartphone production and consumption while encouraging them to consider the economic and social consequences of sustainability measures. Its emphasis on competing priorities supports critical thinking and responsible decision-making. Design and Technology lessons can further explore smartphone lifecycles, recycling, and technological innovations. In Mathematics, students can use the game's decision tree to investigate combinatorics and probability, calculate theoretically possible playthroughs, and explore how random events influence outcomes.

Italy

Triple Balance can be integrated into lower-secondary education (scuola secondaria di primo grado), particularly in Civic Education (Educazione civica), Geography (Geografia), Technology (Tecnologia), and Mathematics (Matematica). It aligns especially well with the Civic Education curriculum's focus on economic development and sustainability (Sviluppo economico e sostenibilità). Students explore the transition to a circular economy while confronting dilemmas involving environmental protection, financial constraints, and public acceptance. Geography and Technology provide opportunities to investigate the environmental footprint of smartphones, resource extraction, recycling, and sustainable product design. In Mathematics, the game offers practical applications of combinatorics and probability through calculating possible playthroughs and analyzing random events.

6 Low-Cost Replication

Apart from a digital device (desktop computer, laptop, tablet, or phone), nothing is needed to play the game as many times as you want. Nothing needs to be installed on the device, the game is accessible via the internet browser of your choice.

For some of the optional activities, it is helpful to use printouts.

It is possible to download an offline version of the game from the repository:

https://github.com/IMAGINARY/triple-balance-phone

The game is open source, and you may create translations and/or adaptations.

7 Worksheets

No worksheets are needed for this activity. For the optional math challenges, printouts are beneficial, though not strictly needed.

Direct links for download:

- Decision tree without results: https://github.com/IMAGINARY/triple-balance-phone/blob/main/educational_material/en/decision_tree_no_results.png
- Decision overview: https://github.com/IMAGINARY/triple-balance-phone/blob/main/educational_material/en/overview_decisions.png
- Find additional information in this folder: https://github.com/IMAGINARY/triple-balance-phone/tree/main/educational_material (there is a subfolder for each available language)